

Train, Validate, Test

Basic Neural Networks

(and some Deep Network Fundamentals)

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Classical Types of Learning

- Supervised Learning - learn to predict an output when given an input vector
- Unsupervised Learning - discover a good internal representation of the input
- Reinforcement Learning - learn to select an action to maximize the expectation of future rewards (payoff)
- Semi-supervised Learning - learn with few labelled examples and many unlabelled ones

Other Types of Learning

- Self-supervised Learning - learn with targets induced by a prior on the unlabelled training data
- Active Learning - learn by seeking guidance from human or oracle when needed (iterative semi-supervised learning)
- Continual Learning - learn new tasks/classes sequentially (iterative supervised/unsupervised learning)
- Online learning - learning one example at a time sequentially (iterative supervised learning)

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 - Multiclass classification - target is one of the k classes
 - Multilabel classification - target is some number of the k classes
 - In both cases, the machine is a function $f : \mathbb{R}^n \rightarrow \{1, \dots, k\}$.

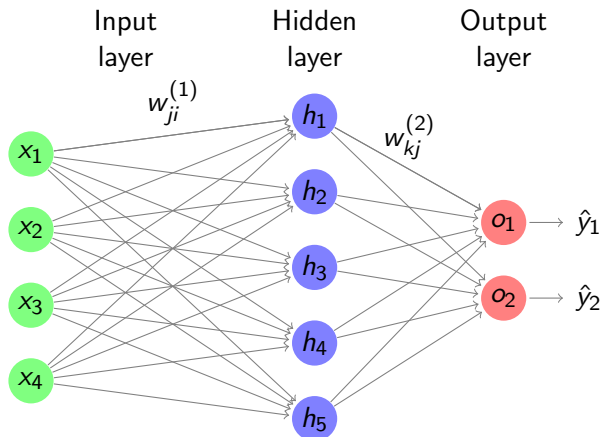
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- It is most common for both types of algorithms to actually learn $\hat{f} : \mathbb{R}^n \rightarrow \mathbb{R}^k$.

How Supervised Learning Typically Works

- Start by choosing a model-class: $\hat{y} = f(\mathbf{x}; \mathbf{W})$ where the model-class f is a way of using some numerical parameters, $\mathbf{W}$, to map each input vector $\mathbf{x}$ to a predicted output $\hat{y}$.
- *Learning* means adjusting the parameters to reduce the discrepancy between the true target output y on each training case and the output $\hat{y}$, predicted by the model.

Let's look at a Multilayer Perceptron (without biases)...



Without loss of generality, we can write the above as:

$$\hat{\mathbf{y}} = g(f(\mathbf{x}; \mathbf{W}^{(1)}); \mathbf{W}^{(2)}) = g(\mathbf{W}^{(2)} f(\mathbf{W}^{(1)} \mathbf{x}))$$

where f and g are activation functions.

Common Activation Functions

- Identity
- Sigmoid (aka Logistic)
- Hyperbolic Tangent ($\tanh$)
- Rectified Linear Unit (ReLU) (aka Threshold Linear)

Final layer activations

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- What form should the final layer function g take?
- It depends on the task (and on the chosen loss function)...
 - regression $\rightarrow$ typically linear
 - binary classification $\rightarrow$ typically Sigmoid
 - multilabel classification $\rightarrow$ typically Sigmoid
 - multiclass classification $\rightarrow$ typically Softmax

$$\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad \forall i = 1, 2, \dots, K$$

- Note that softmax makes reference to all the elements in the output.
- output: positive numbers that sum to 1.
- Note:

$$\begin{aligned} \frac{\partial \text{softmax}(\mathbf{z})_i}{\partial z_i} &= \text{softmax}(z_i)(1 - \text{softmax}(z_i)) \\ \frac{\partial \text{softmax}(\mathbf{z})_i}{\partial z_j} &= \text{softmax}(z_i)(1(i = j) - \text{softmax}(z_j)) \\ &= \text{softmax}(z_i)(\delta_{ij} - \text{softmax}(z_j)) \end{aligned}$$

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- There are many different loss functions we might encounter (MSE, Cross-Entropy, KL-Divergence, huber, L1 (MAE), CTC, Triplet, ...) for different tasks.

The Loss Function (measure of discrepancy)

Recall from Foundations of Machine Learning:

- Mean Squared Error (MSE) loss for a single data point is given by

$$\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y}) = \sum_i (\hat{y}_i - y_i)^2 = (\hat{\mathbf{y}} - \mathbf{y})^\top (\hat{\mathbf{y}} - \mathbf{y})$$

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- $\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y})$ is the predominant choice for regression problems with linear activation in the last layer
- For a classification problem with Softmax or Sigmoidal activations MSE can cause slow learning
 - Gradients of ℓ_{MSE} are proportional to the difference in target and predicted value, multiplied by the gradient of the activation function
 - **The Cross-Entropy loss function is generally a better choice in this case**

Binary Cross-Entropy

For the binary classification case:

$$\ell_{BCE}(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

- The cross-entropy loss function is non-negative, $\ell_{BCE} > 0$
- $\ell_{BCE} \approx 0$ when the prediction and targets are equal (i.e. $y = 0$ and $\hat{y} = 0$ or when $y = 1$ and $\hat{y} = 1$)
- With Sigmoidal final layer, $\frac{\partial \ell_{BCE}}{\partial \mathbf{w}_i^{(2)}}$ is proportional to just the error in the output ($\hat{y} - y$) and therefore, the larger the error, the faster the network will learn!
- Note that the BCE is the negative log likelihood of the Bernoulli Distribution

Binary Cross-Entropy — Intuition

- The cross-entropy can be thought of as a **measure of surprise**.
- Given some input x_i , we can think of $\hat{y}_i$ as the estimated probability that x_i belongs to class 1, and $1 - \hat{y}_i$ is the estimated probability that it belongs to class 0.
- Note the extreme case of infinite cross-entropy, if your model believes that a class has 0 probability of occurrence, and yet the class appears in the data, the 'surprise' of your model will be infinitely great.

Binary Cross-Entropy for multiple labels

In the case of multi-label classification with a network with multiple sigmoidal outputs you just sum the BCE over the outputs:

$$\ell_{BCE} = - \sum_{k=1}^K [y_k \log(\hat{y}_k) + (1 - y_k) \log(1 - \hat{y}_k)]$$

where K is the number of classes of the classification problem, $\hat{y} \in \mathbb{R}^K$.

Numerical Stability: The Log-Sum-Exp trick

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- Consider what might happen early in training when the model might confidently predict a positive example as negative
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- To tackle this problem implementations usually combine the sigmoid computation and BCE into a single loss function that you would apply to a network with linear outputs (e.g. `BCEWithLogitsLoss`).
- Internally, a trick called 'log-sum-exp' is used to *shift* the centre of an exponential sum so that only numerical underflow can potentially happen, rather than overflow
- - Ultimately this means you'll always get a numerically reasonable result (and will avoid NaNs and Infs originating from this point).

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- In such a case, the obvious loss function is the *negative log likelihood* of the Categorical distribution (aka Multinoulli, Generalised Bernoulli, Multinomial with one sample)
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Log-Sum-Exp can be used for better numerical stability. PyTorch combines LogSoftmax with NLL in one loss and calls this “Categorical Cross-Entropy” (so you would use this with a **linear output layer**)

Reminder: Gradient Descent

- Define total loss as $\mathcal{L} = \sum_{(\mathbf{x}, y) \in \mathbf{D}} \ell(g(\mathbf{x}, \boldsymbol{\theta}), y)$ for some loss function ℓ , dataset $\mathbf{D}$ and model g with learnable parameters $\boldsymbol{\theta}$.
- Define how many passes over the data to make (each one known as an Epoch)
- Define a learning rate λ

Gradient Descent updates the parameters $\boldsymbol{\theta}$ by moving them in the direction of the negative gradient with respect to the **total loss** $\mathcal{L}$ by the learning rate λ multiplied by the gradient:

for each Epoch:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \lambda \nabla_{\boldsymbol{\theta}} \mathcal{L}$$

Reminder: Stochastic Gradient Descent

- Define loss function ℓ , dataset $\mathbf{D}$ and model g with learnable parameters θ .
- Define how many passes over the data to make (each one known as an Epoch)
- Define a learning rate λ

Stochastic Gradient Descent updates the parameters θ by moving them in the direction of the negative gradient with respect to the loss of a **single item** ℓ by the learning rate λ multiplied by the gradient:

```
for each Epoch:
    for each  $(\mathbf{x}, y) \in \mathbf{D}$ :
         $\theta \leftarrow \theta - \lambda \nabla_{\theta} \ell$ 
```

A Quick Introduction to Tensors

Broadly speaking a tensor is defined as a linear mapping between sets of algebraic objects¹.

A tensor T can be thought of as a generalization of scalars, vectors and matrices to a single algebraic object.

We can just think of this as a multidimensional array².

- A $0D$ tensor is a scalar
- A $1D$ tensor is a vector
- A $2D$ tensor is a matrix
- A $3D$ tensor can be thought of as a vector of identically sized matrices
- A $4D$ tensor can be thought of as a matrix of identically sized matrices or a sequence of $3D$ tensors
- ...

¹This statement is always entirely true

²This statement will upset mathematicians and physicists because its not always true for them (but it is for us!).

Operations on Tensors in PyTorch

- PyTorch lets you do all the standard matrix operations on 2D tensors
 - including important things you might not yet have seen like the **hadamard product** of two $N \times M$ matrices: $\mathbf{A} \odot \mathbf{B}$)
- You can do element-wise add/divide/subtract/multiply to ND-tensors
 - and even apply scalar functions element-wise (log, sin, exp, ...)
- you can slice, reshape, and *even index a single element* (**generally don't do that!**)
- PyTorch often lets you *broadcast* operations (just like in numpy)
 - if a PyTorch operation supports broadcast, then its Tensor arguments can be automatically expanded to be of equal sizes (without making copies of the data).³

³Important - read and understand this after the lab:
<https://pytorch.org/docs/stable/notes/broadcasting.html>

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- Broadcasting is used to apply operations/functions to all the samples in the batch tensor *in parallel* to compute a loss vector
- the loss vector is summed/averaged using a *vectorised* method (e.g. `.sum()`)

PyTorch Tensor 101:

<https://colab.research.google.com/gist/jonhare/d98813b2224dddbb234d2031510878e1/notebook.ipynb>

Watch and understand this:

<https://southampton.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=c62809ad-af4d-4c7f-89e1-b26f00f85cd9>